Hazard Rate

$H(x) = -\log(1-F(x))$

$r(x) = H'(x) = \dfrac{f(x)}{1-F(x)}$

Exp. R.V.

$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & o.w. \end{cases}$

$F(x) = 1 - e^{-\lambda x}, \quad x \geq 0$

$E(X) = 1/\lambda$, Median $= \dfrac{\ln 2}{\lambda}$,

$E(X^2) = 2/\lambda^2$

$var(X) = 1/\lambda^2$

Gamma(λ, t)

$\Gamma(\lambda, t), \lambda > 0.$

$f(x) = \dfrac{1}{\Gamma(t)} \lambda^t x^{t-1} e^{-\lambda x}, \quad x \geq 0$

$E(X) = t/\lambda$

$Var(X) = t/\lambda^2$

$\Gamma(t) = \int_0^\infty x^{t-1} e^{-x} dx$

$\Gamma(t+1) = t! = t\Gamma(t)$

$\Gamma(1/2) = \sqrt{\pi}$

$t = 1 \Rightarrow \Gamma(\lambda, t) \sim \exp(\lambda)$

$\lambda = 1/2, t = d/2 \Rightarrow \Gamma(\lambda, t) \sim \chi^2(d)$

Weibull R.V.

Weibull(α, β)

$F(x) = 1 - e^{-\alpha x^\beta}$

$f(x) = \alpha\beta x^{\beta-1} e^{-\alpha x^\beta} \quad \alpha, \beta > 0$

if $\beta = 1$, Weibull$(\alpha, 1) \sim \exp(\alpha)$

$E(X^k) = \dfrac{k\Gamma(k/\beta)}{\beta \alpha^{k/\beta}}$

$E(X) = \dfrac{\Gamma(1/\beta)}{\beta \alpha^{1/\beta}}$

$Var(X) = \dfrac{\Gamma(2/\beta + 1) - \Gamma(1/\beta)^2}{\beta \alpha^{2/\beta}}$

Normal Distribution $N(\mu, \sigma^2)$

$f(x) = \dfrac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$

$E(X) = \mu$, $E(X^2) = \mu^2 + \sigma^2$

$Var(X) = \sigma^2$

$\mu = 0, \sigma^2 = 1 \Leftrightarrow N(0,1)$ (Stand. nor. distn)

Uniform distn

$f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & o.w. \end{cases}$

$E(X) = \dfrac{(a+b)}{2}$

$Var(X) = (b-a)^2/12$

Bivariate Normal Distn.

$f(x,y) = \dfrac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}(x^2 - 2\rho xy + y^2)} = \dfrac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}(x-\rho y)^2} e^{-\frac{y^2}{2}}$

$-\infty < x, y < \infty, \quad -1 < \rho < 1, \quad f_X(x) = \dfrac{1}{\sqrt{2\pi}} e^{-x^2/2} \rightarrow X, Y \sim N(0,1)$

$E(X) = E(Y) = 0, \quad Var(X) = Var(Y) = 1, \quad Cov(X,Y) = \rho = E(XY)$

Beta distn

$f(x) = \dfrac{x^{a-1}(1-x)^{b-1}}{\int_0^1 x^{a-1}(1-x)^{b-1} dx} \quad 0 \leq x \leq 1, \quad B(a,b) = \int_0^1 x^{a-1}(1-x)^{b-1} dx$

$E(X) = \dfrac{a}{a+b}, \quad Var(X) = \dfrac{a(a+1)}{(a+b)(a+b+1)}$

* $a = b = 1$, $X \sim$ uniform $[0,1]$

Conditional Distribution

$$F_{Y/X}(y/x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

$$E[Y/X] = \int y f_{Y/X}(y/x)\, dy$$

$$\psi(x) = E(Y/X)$$

$$E(\psi(x)) = E_X[E_{Y/X}[Y/X]] = E(Y)$$

$$E[X^i Y^j] = E_X[X^i E_{Y/X}[Y^j]]$$

Skewness

$$skw(X) = \frac{E((X-m_1)^3)}{\sigma^3}$$

Kurtosis

$$Kur(X) = \frac{E((X-m_1)^4)}{\sigma^4}$$

Thm If $x_1(y), x_2(y), \ldots, x_n(y)$ such that $g(x_k(y)) = y$, $g'(x_k(y)) \neq 0$.

$$h(y) = \begin{cases} \sum_{k=1}^{n} f(x_k(y)) \left| g'(x_k(y)) \right|^{-1} & ; x > 0 \\ 0 & o.w. \end{cases}$$

$$\omega(y_1, y_2, \ldots, y_n) = |J| \cdot f(h_1(y_1, y_2, \ldots, y_n)) \cdot h_2(y_1, \ldots, y_n) \ldots h_n$$

Probability Generating function

$$G_X(z) = E[z^X] = \sum_{x=0}^{\infty} f(x) \cdot z^x \quad |z| \leq 1$$

> $X \sim$ Bernoulli$(p) \Rightarrow G_X(z) = zp + (1-p) = pz + q$

> $X \sim bin(n, p) \Rightarrow G_X(z) = \sum_{k=0}^{n} {}^nC_k p^k q^{n-k} z^k = (pz+q)^n$

> $X \sim$ Poisson$(\lambda) \Rightarrow G_X(z) = E[z^X] = e^{\lambda(z-1)}$

> $X \sim$ Geometric$(p) \Rightarrow f(x) = (1-p)^{x-1} p \Rightarrow G_X(z) = \frac{pz}{1-qz}$

Properties.

(i) $E(X) = G'(1)$, (ii) $E(X(X-1)) = G''(1)$, (iii) $var(X) = G''(1) - G'(1)$

(iv) $X, Y \rightarrow$ independent $\Rightarrow G_{X+Y}(z) = G_X(z) G_Y(z)$

Moment gen. f^n $(M: \mathbb{R} \to [0, \infty))$

$$M_X(t) = E[e^{tX}] = \begin{cases} \sum_x e^{tx} f(x) \\ \int_{-\infty}^{+\infty} e^{tx} f(x)\,dx \end{cases}$$

$$M_X(t) = \int_{-\infty}^{+\infty} e^{tx} f(x)\,dx$$

$$= \int_{-\infty}^{+\infty} \left(1 + tx + \frac{t^2x^2}{2!} + \dots\right) f(x)\,dx = 1 + \mu_1' t + \frac{\mu_2'}{2!} t^2 + \frac{\mu_3'}{3!} t^3 + \dots$$

$$\left[\mu_i' = E[X^i]\right]$$

$M_{X+Y}(t) = M_X(t) M_Y$

$\rightarrow X \sim \exp(\lambda) \Rightarrow M_X(t) = \lambda/(\lambda - t)$ $\{t < \lambda\}$

$\rightarrow X \sim N(\mu, \sigma^2) \Rightarrow M_X(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$

$\rightarrow E[X^k] = M^k(0) \Rightarrow E(X) = M'(0)$
$E(X^2) = M''(0)$

~~Chebyshev's Ineq~~

$P\{h(x) \geq \varepsilon\} \leq \frac{E(h)}{\varepsilon}$

Markov's inequality

$P(|X| \geq k) \leq \frac{E[|X|^r]}{k^r}$

Chebyshev's Ineq.

$h(X) = (X - \mu)^2$, $\varepsilon = k^2\sigma^2$
$\mu = E(X)$, $\sigma^2 = Var(X)$
$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$

$\Rightarrow P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$

Strong law of large no.

$\bar{X}_n \to \mu$ almost surely when $n \to \infty$

$P\{\lim_{n\to\infty} \bar{X}_n = \mu\} = 1$

Convergence

$X_1, X_2, \dots \infty \to$ independent & iid r.v.

$E(X_1) = E(X_2) = \dots = \mu$
$Var(X_1) = Var(X_2) = \dots = \sigma^2$

$\bar{X}_n = \frac{\sum X_i}{n}$
↓
Sample avg./mean

Weak Law of large no.

As $n \to \infty$, $\lim_{n\to\infty} P\{|\bar{X}_n - \mu| \geq \varepsilon\} = 0$

for any arbitary $\varepsilon > 0$, $P\{|\bar{X}_n - \mu| < \varepsilon\} \to 1$ as $n \to \infty$

Joint probability density f^n

$G_{X_1 X_2}(z_1, z_2) = E[z_1^{X_1} z_2^{X_2}]$

Joint moment generating f^n

$M_{X,Y}(s,t) = E[e^{sX} e^{tY}]$

if X_1 & $X_2 \to$ indep. $\Rightarrow G_{X_1,X_2}(z_1, z_2) = G_{X_1}(z_1) \cdot G_{X_2}(z_2)$

$X, Y \to$ indep. $M_{X,Y}(s,t) = M_X(s) \cdot M_Y(t)$

Central limit theorem

infinite sample,

$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ | if $n \to \infty$, $z \sim N(0,1)$

$K(t) = \ln E(e^{tx}) = \ln(M(t)) = \ln\left(1 + \mu_1' t + \frac{\mu_2' t^2}{2!} + \dots\right)$

$K(t) = \frac{k_1 t}{1!} + \frac{k_2 t^2}{2!} + \frac{k_3 t^3}{3!} + \dots$

$k_1 = E(X) = \mu_1'$

$k_2 = Var(X) = E((X-\mu)^2) = \sigma^2$

$k_3 = E[(X-\mu_1')^3] =$ third central moment $= \mu^3$.

$k_4 = E[(X-\mu_1')^4] - 3(E[(X-\mu_1')^2])^2 = \mu_4 - 3\sigma^4$.

$M(t) = e^{K(t)}$

$\Rightarrow 1 + \sum_{i=1}^{\infty} \frac{\mu_i'}{i!} t^i = e^{\left[k_1 \frac{t}{1!} + \frac{k_2 t^2}{2!} + \dots\right]}$

$\mu_1' = k_1$, $\mu_2' = k_2 + k_1^2 \Rightarrow \mu_3' = k_3 + 3k_1 k_2 + k_1^3$

$\mu_4' = k_4 + 4k_3 k_1 + 3k_2^2 + 6k_2 k_1^2 + k_1^4$

Central Moment Generating f^n

$C(t) = E[e^{t(x-\mu_1')}] = E[e^{t(x-\mu)}] = e^{-\mu t} M(t) = e^{K(t) - \mu t}$

$\mu_n = C^n(0)$

$\mu_1 = 0$, $\mu_2 = k_2$, $\mu_3 = k_3$, $\mu_4 = k_4 + 3k_2^2$

$\mu_1 = C'(t) = e^{(K(t) - \mu t)}[K_1'(0) - \mu] = 0$.

(I) $n > 1$, $K_n(x+c) = K_n(x)$

$n = 1$, $K_1(x+c) = K_1(x) + c$

(II) $X_1, X_2, \dots \to$ indep.

$K_n(X_1 + X_2 + \dots + X_m) = K(X_1) + K(X_2) + \dots + K_m(X_m)$.

III) $K_n(CX) = C^n K_n(x)$.

Chi-Sq. Dist^n

$f(x) = \begin{cases} \frac{1}{2^{w/2}\Gamma(w/2)} x^{\frac{w-2}{2}} e^{-x/2} & x > 0 \\ 0 & \text{o.w.} \end{cases} \Rightarrow M_x(t) = (1-2t)^{-v/2}$

$E(X) = v$, $Var(X) = 2v$

$X \sim N(0,1) \Rightarrow X^2 \sim \chi^2(1)$

- $\bar{X}$ & $S^2 \to$ sample mean & variance of random sample of size n.

mean μ, variance σ^2,

$\bar{X}, S^2 \to$ Independent.

$(n-1)\frac{S^2}{\sigma^2} \sim \chi^2(n-1)$

Student t dist^n

$Z \sim N(0,1)$, $Y \sim \chi^2(v) \Rightarrow T = \frac{Z}{\sqrt{Y/v}} \sim t(v)$

$$f(t) = \frac{\Gamma((v+1)/2)}{\sqrt{\pi v}\,\Gamma(v/2)} (1 + t^2/v), \quad -\infty < t < \infty, \quad \boxed{P(t \geq t_{\alpha,v})}$$

$$\boxed{T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)}$$

$\bar{X} \sim N(\mu, \sigma^2/n)$

$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$

Student F dist^n

$$F = \frac{U/v_1}{V/v_2}$$

$U \sim \chi^2(v_1)$

$V \sim \chi^2(v_2)$

$$g(f) = \begin{cases} \frac{\Gamma\left(\frac{v_1+v_2}{2}\right)}{\Gamma(v_1/2)\Gamma(v_2/2)} (v_1/v_2)^{v_1/2} f^{v_1/2 - 1} \left(1 + \frac{v_1 f}{v_2}\right)^{-\frac{(v_1+v_2)}{2}}, & f > 0 \\ 0 & \text{o.w.} \end{cases}$$

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} = \frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2} \Rightarrow F \sim F(n_1 - 1, n_2 - 1)$$

$$\frac{(n_1-1)S_1^2}{\sigma_1^2} \sim \chi^2(n_1-1), \quad \frac{(n_2-1)S_2^2}{\sigma_2^2} \sim \chi^2(n_2-1)$$

* $(1-x)^{-n} = \sum_{k=0}^{\infty} \binom{n+k-1}{k} x^k$

* $X \sim \chi(v)$

$$M_X(t) = \begin{cases} (1-2t)^{-v/2}, & t < 1/2 \\ \text{DNE} & t \geq 1/2 \end{cases}$$